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End Semester Examination 2025

Name of the Course: B.Tech. Biotechnology

Name of the Paper: Nanobiotechnology

Time: 3 Hours

Semester: V

Paper Code: **TBT506**

Maximum Marks: 100

Note:-

- (iv) All questions are compulsory.
- (v) Answer any two sub questions among a, b & c in each main question.
- (vi) Each sub question carries 10 marks.

Q1	(20 marks)	
(a)	Briefly discuss the effect of nano-dimensions on electrical, magnetic, and optical properties of materials.	CO1
(b)	What is significance of nano-domain? Briefly discuss about different forces involved at nano-domain.	
(c)	Give an account of the discovery of CNTs and briefly explain CVD and Laser Ablation methods of synthesizing CNTs.	
Q2	(20 marks)	
(a)	What is the principle of X-ray diffraction ? Elaborate on working and applications of X-ray diffraction in nanotechnology.	CO2
(b)	What do you understand by green synthesis? How nanoparticles can be synthesized using plant metabolites and microbes?	
(c)	Briefly describe co-precipitation and microemulsion techniques of fabricating nanomaterials along with their general mechanism, reaction characteristics and advantages.	
Q3	(20 marks)	
(a)	Define nanobiosensors and briefly discuss about electrochemical and optical transducers.	CO3
(b)	Briefly discuss the various types of polymeric drug delivery systems.	
(c)	What is quantum dots technology? Discuss their classification, and applications in bio-imaging and bio-sensors.	
Q4	(20 marks)	
(a)	Discuss the role and mechanisms of metal and metal oxide nanoparticles in countering infectious diseases.	CO4
(b)	How nanotechnology can be used for treating hypertension and cardio-vascular disease?	
(c)	What are magnetic nanoparticles and how they can be used for tumor targeting and diagnostic? Briefly discuss the process of Magnetic fluid hyperthermia .	
Q5	(20 marks)	
(a)	Comment on the role of nanotechnology in detection of food pathogens.	CO5
(b)	Explain the process of photocatalytic degradation of organic pollutants using nanomaterials.	
(c)	How nanotechnology can help in food packaging and preservation?	

